

Harjot Singh



Aerocity, Mohali, Punjab



6280741707



www.linkedin.com/in/harjot-singh-64b9951aa



[GitHub](#)

✉ Harjot2797@gmail.com

PROFILE

I am an dedicated Data Science with a strong background in data management, analysis, and optimization. I have developed advanced skills in designing and managing complex databases, writing efficient queries, and performing large-scale data migrations. My proficiency in SQL is complemented by my knowledge of Machine Learning, enabling me to implement data-driven solutions that provide valuable insights for predictive analytics and automation.

TECHNICAL STACK

MYSQL	PostGRE SQL	Database Designing
Python	Machine Learning	Deep Learning
Data Visualization	Statistics	Power BI

INDUSTRIAL PROJECTS

Object Detection Platform using CNN and Python in Jupyter Notebook

[GitHub](#)

- **Concept of Object Detection:** Object detection identifies and localizes multiple objects in an image or video using bounding boxes. It differs from image classification by providing object positions along with labels. This technique is essential for applications like surveillance, autonomous vehicles, and robotics.
- **CNN-Based Object Detection Models:** Convolutional Neural Networks (CNNs) power modern object detection by extracting spatial features. Popular models include YOLO (fast), Faster R-CNN (accurate), and SSD (balanced). These models process images efficiently to detect objects in real time.
- **Applications of Object Detection:** Object detection is widely used in smart surveillance, self-driving cars, and medical imaging. It enables facial recognition, crowd monitoring, and license plate detection. Retail analytics also use it for inventory tracking and customer behavior analysis.
- **Data Processing and Model Training:** Models are trained on datasets like COCO and Pascal VOC, using labeled images with bounding boxes. Feature extraction and region proposal methods improve accuracy. Data augmentation, transfer learning, and optimization techniques enhance performance.
- **Implementation and Deployment** Object detection is implemented using Python libraries like TensorFlow, PyTorch, and OpenCV. Deployment can be done on cloud platforms, edge devices, or mobile applications. Optimizations like quantization and model pruning improve real-time performance.

Stock Forecasting Technology Used: Python, Machine Learning using (LSTM)

[GitHub](#)

- **Data Collection and Preprocessing** Stock data is collected from financial APIs (like Alpha Vantage, Yahoo Finance) and stored in a database using SQL. The data is then cleaned, normalized, and transformed for use in forecasting models.
- **Model Building and Training** Machine learning models like LSTM (Long Short-Term Memory) or ARIMA are used for predicting stock prices. These models learn from historical price data to forecast future stock movements for the next 10 days.
- **Evaluation and Prediction** The performance of the model is evaluated using metrics like Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). Once the model performs well, it is used to predict stock prices for the upcoming days, with results displayed via visualization tools.

Second-Hand Car Price Prediction using Machine Learning[GitHub](#)

Technology used: Python, Numpy, Pandas, Statistics and Machine learning

- Collected data on second-hand cars, including features like model, year, mileage, and price.
- Cleaned and prepared the data by handling missing values, encoding categories, and scaling numerical features.
- Built and trained machine learning models such as Linear Regression, Random Forest, and XGBoost to predict car prices.
- Deployed the model for users to input car details and receive price estimates, while visualizing trends with Power BI.

EDUCATION & CERTIFICATIONS		EXTRACURRICULAR ACTIVITIES	
Infant Jesus and Convert School,Mohali	2019	Cricket	
CBSE			
Shaheed Udham Singh,Tangori	2023	Reading and Writing	
Diploma		Meditation	
Chandigarh University	2025		
Bachelor of Engineering (CSE)			
Mysql and PLSQL Certification	2024		
Theta Academy			